

CUBClustR: Clustering Rating data within the CUB framework

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April 25, 2026

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1 Introduction

This document provides the comprehensive technical documentation for the implementation of the **MLC-CUB** (Multivariate Latent Class CUB) model in R. The framework allows segmenting ordinal data (ratings) by accounting for both the *feeling* process (reasoned choice) and the *uncertainty* of the respondent, grouping subjects into K latent clusters.

The estimation relies on the Expectation-Maximization (EM) algorithm, deeply optimized via a **C++** backend (**Rcpp**) to ensure high computational performance.

To use the library, keep the file `CUBClustR_functions.R` and the file `engine.cpp` in the same folder. Upload the function with the R command `source`.

2 Function Reference

2.1 fit_MLCCUB

Description: Estimates the parameters of an MLC-CUB model using the C++ accelerated EM algorithm.

Usage:

```
1 fit_MLCCUB(data, m, K, tol = 1e-5, EM_iter = 10, max_iter = 1500)
```

Arguments (Inputs):

- **data:** A numeric matrix or dataframe containing the ordinal responses. Rows represent subjects (N) and columns represent items (J).
- **m:** A numeric vector specifying the maximum number of response categories for each item.
- **K:** Integer. The number of latent classes (clusters) to estimate.
- **tol:** Numeric. Tolerance for the log-likelihood convergence. (*Default: 1e-5*).
- **EM_iter:** Integer. Number of random initializations to avoid local maxima. (*Default: 10*).
- **max_iter:** Integer. Maximum number of EM iterations. (*Default: 1500*).

Value (Outputs): A list containing:

- **pi, xi, omega:** Estimated uncertainty, feeling, and weight parameters.
- **LL, class:** Final Log-Likelihood and Maximum A Posteriori (MAP) cluster assignments.

2.2 compare_MLCCUB_models

Description: Fits multiple MLC-CUB models for a sequence of latent classes (K) and compares them to facilitate model selection using the Bayesian Information Criterion (BIC).

Usage:

```
1 compare_MLCCUB_models(data, m, K_vector, ...)
```

Arguments (Inputs):

- **data:** A numeric matrix or dataframe containing the observed ordinal ratings.
- **m:** A numeric vector specifying the maximum categories for each item.
- **K_vector:** An integer vector containing the range of clusters to be tested (e.g., 1:6).
- **...:** Additional arguments passed directly to `fit_MLCCUB` (e.g., `EM_iter`, `tol`).

Value (Outputs): A list containing:

- **all_models:** A named list containing all the successfully fitted model objects.

- **summary:** A `data.frame` summarizing the results for each K , including Log-Likelihood, number of parameters, and BIC values.
- **best_k:** Integer. The number of clusters that yielded the lowest BIC.
- **best_model:** The complete model object corresponding to the optimal BIC.

2.3 check_identifiability

Description: Performs a non-parametric Bootstrap analysis to assess the stability and identifiability of the latent clusters. It calculates the Adjusted Rand Index (ARI) between assignments across different bootstrap samples.

Usage:

```
1 check_identifiability(data, m, K, n_boot = 100, n_cores = 1,
2                       plot_hist = TRUE, seed = 250535, ...)
```

Arguments (Inputs):

- **data:** The original dataset used for the analysis.
- **m:** Numeric vector of maximum categories per item.
- **K:** Integer. The number of clusters to test for stability.
- **n_boot:** Integer. The number of bootstrap samples to extract. (*Default: 100*).
- **n_cores:** Integer. Number of CPU cores for parallel computation. (*Default: 1*).
- **plot_hist:** Logical. If `TRUE`, generates the ARI distribution histogram.

Value (Outputs): A list containing:

- **ARI_matrix:** A symmetric matrix containing pairwise ARI scores between bootstrap iterations.
- **ARI_values:** A numeric vector of the pairwise ARI scores (flattened).
- **plot:** A `ggplot2` object showing the ARI distribution.
- **raw_clusters:** A list containing subject assignments for every bootstrap run.

2.4 calculate_WAD

Description: Computes the Weighted Average Dissimilarity (WAD) index and provides granular diagnostic metrics to evaluate the absolute goodness of fit.

Usage:

```
1 calculate_WAD(model_obj, data, m)
```

Arguments (Inputs):

- **model_obj:** A fitted model object.
- **data:** The original dataset. The function internally ensures compatibility via matrix conversion and index-matching checks.

- **m**: The vector of maximum categories.

Value (Outputs): A diagnostic list containing:

- **WAD**: Global fit index in $[0, 1]$. Values closer to 0 indicate excellent fit.
- **Cluster_Dissimilarity**: Unweighted average dissimilarity for each cluster.
- **Item_Dissimilarity**: Average dissimilarity per item, weighted by cluster sizes (ω).
- **Matrix_Dissimilarity**: The full $K \times J$ dissimilarity matrix.

2.5 Plotting Functions

2.5.1 plot_MLCCUB_clusters

Description: Generates a 2D parameter space plot (Feeling vs Uncertainty) for every item, grouped by cluster.

- **cluster_order**: Optional vector to manually reorder clusters in the legend.
- **custom_colors**: Character vector for custom HEX color palettes.

2.5.2 plot_CUB_fit

Description: Creates a multi-panel facet grid ($Clusters \times Items$) comparing observed relative frequencies (solid black segments) with theoretical CUB distributions (dashed lines).

Usage:

```
1 plot_CUB_fit(model_obj, data, m, cluster_order = NULL, line_color = "darkblue")
```

Customization:

- **line_color**: Change the HEX string to customize the theoretical probability line (e.g., "#E74C3C" for red). *Default: Dark Blue.*
- **Robustness**: The function automatically handles missing values and prevents "subscript out of bounds" errors by verifying row consistency between the model and the data.